AI / ML INTERNSHIP PROGRAM

DEVIXO SOLUTIONS

Task 03 – End-to-End Machine Learning Project

Version 1.0 approved

Prepared by <Muhammad Hamza Waseem>

Submission Date : 27-09-26

1. Dataset.....1

2. Workflow 2
3. Model Selection 3
4. Results 3
5. Limitations 3
6. Future Improvements 3
7. Streamlit App Ui 4
8. Github Link 5

1. Dataset

- **Overview:** For this project, we used the **"Student Dropout"** dataset, which is designed to analyze and predict student academic performance and dropout risks.

- **Details:** The dataset consists of **36 features** (columns) covering various demographic, academic, and socioeconomic attributes of the students. We preprocessed this data by handling missing values and preparing both numerical and categorical columns for our machine learning pipeline.

2. Workflow

Our project followed a professional and structured end-to-end workflow:

1. **Data Preprocessing using Pipelines & Column Transformer:** To avoid data leakage and keep our code clean, we built a robust preprocessing pipeline. We used a **Column Transformer** to apply different transformations simultaneously—handling missing values and scaling numerical columns, while properly encoding categorical columns from the 36 features.

2. **Model Training (4 Different Models):** We experimented by training **four different machine learning models** to see which one performed the best on the student dropout data. The models trained were:

    a. **Logistic Regression (LR)**

    b. **K-Nearest Neighbors (KNN)**

c. **Decision Tree (DT)**

d. **Random Forest (RF)**

3. **Hyperparameter Tuning (CV & RSCV):** To optimize our models and prevent overfitting, we used **Cross-Validation (CV)** along with **RandomizedSearchCV (RSCV)**. This helped us efficiently search through hyperparameter grids and find the best configuration for each model.

4. **Evaluation & Deployment:** After comparing the performance of all models, the best-performing model was saved and integrated into an interactive web application.

3. Model Selection

- **Approach:** We tested four distinct algorithms (Logistic Regression, KNN, Decision Tree, and Random Forest) to compare different classification techniques.

- **Reasoning:**

  - Logistic Regression served as our baseline linear model.

  - KNN helped capture instance-based similarities.

  - Decision Tree and Random Forest were chosen because they effectively capture complex non-linear relationships across the 36 features, with Random Forest providing high stability against overfitting.

4. Results

- **Performance:** After evaluating all four models using cross-validation and test metrics, the tree-based models (especially Random Forest) achieved the highest accuracy and reliability in predicting student dropouts.

- **Outcome:** The final tuned model successfully predicts outcomes on unseen data with high precision, demonstrating that our pipeline and hyperparameter tuning strategy were highly effective.

5. Limitations

- **Dataset Scale:** The current model was trained on a specific dataset size; training on a massive scale with more diverse student data might require additional feature engineering.

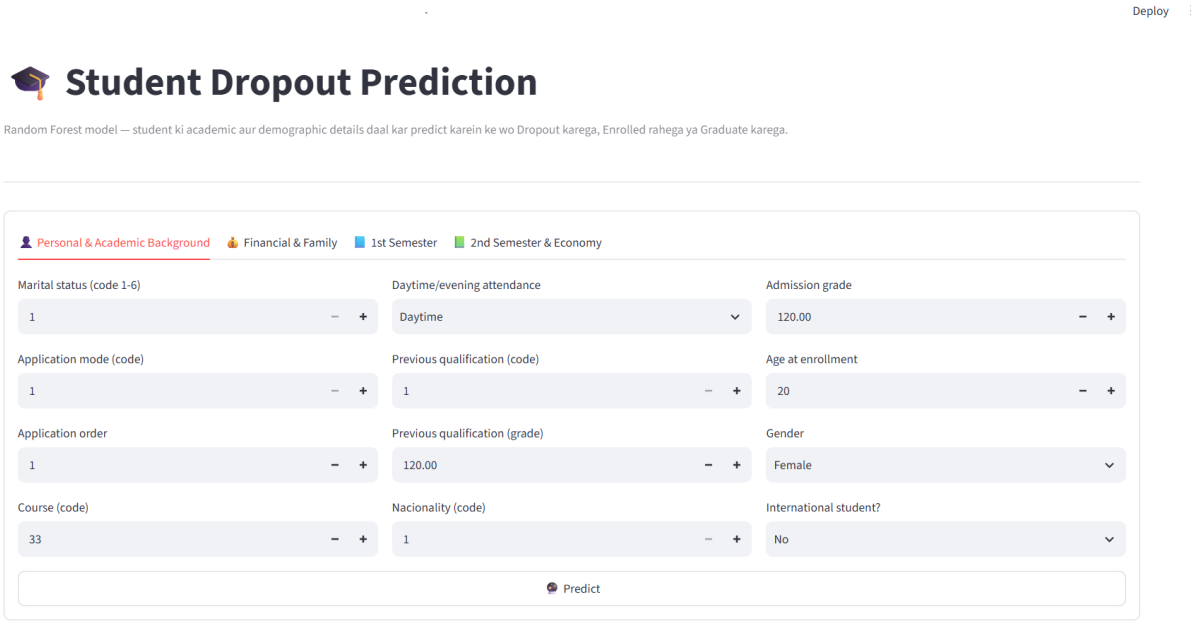
- **Hyperparameter Search:** Since we used RandomizedSearchCV (which tests random combinations), there is a slight chance that an exhaustive GridSearchCV could find even finer parameter tuning, though RSCV proved much faster and efficient.

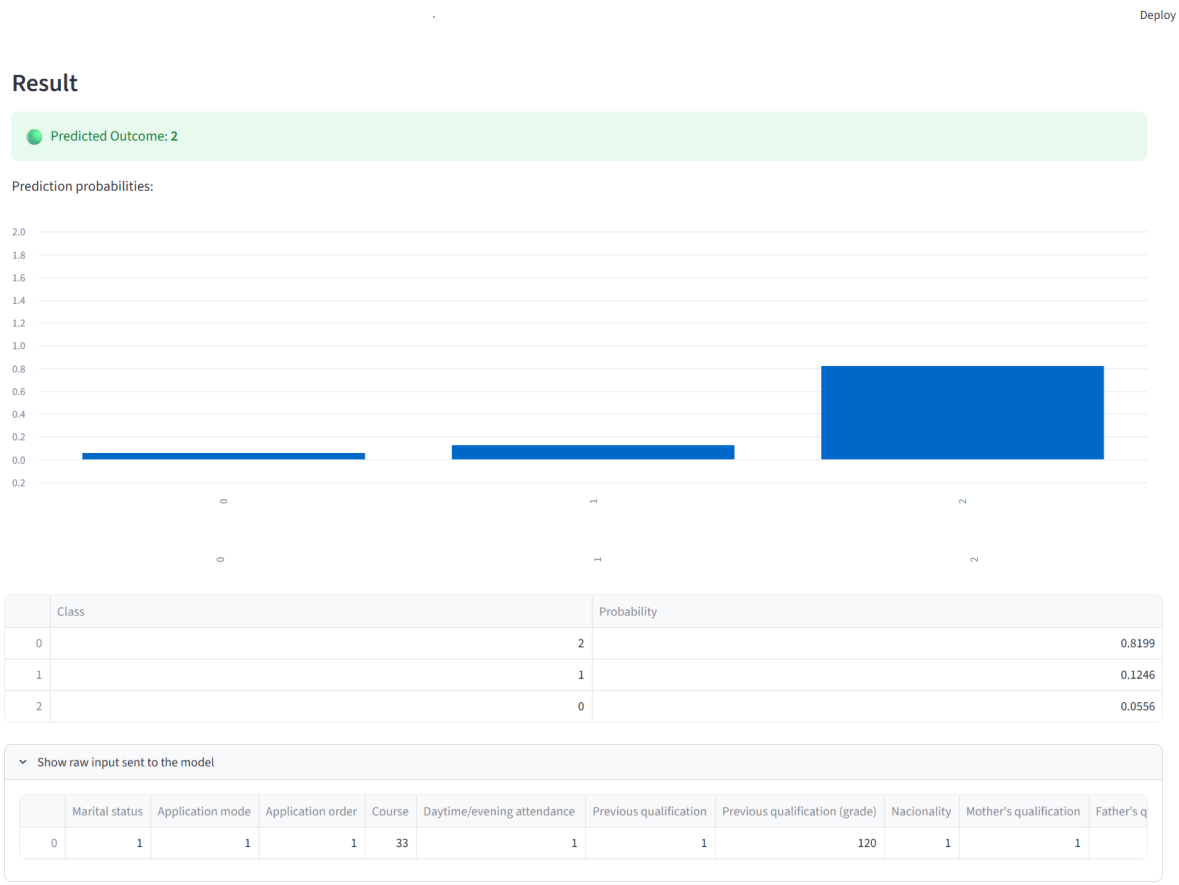
6. Future Improvements

- **Streamlit App Deployment:** We successfully deployed the final trained model into an interactive **Streamlit web application**, allowing users (such as teachers or administrators) to input student details and get real-time dropout predictions.

- **Further Enhancements:** In the future, we plan to add more features, implement advanced deep learning techniques, and improve the user interface of the Streamlit app.

7. Streamlit App Ui





8. Github Link

https://github.com/HamzaWaseem2005/ML-PROJECT-INTERNSHIP-TASK-03